Software Requirements Specification (SRS) Template

Software Requirements Specification (SRS)
Project: Academic Performance Analytics System
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Revision history

Version	Date	Author	Change summary	Approval
1.0	18-08-2025	Instructor	SRS with diagrams embedded	

Approvals

Version	Date	Author	Change Summary	Approval
1.0	12-10-2026	Team 05	Initial SRS prepared for Academic Performance Analytics System, including requirements, security specifications, UML diagrams, and RTM.	Pending

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional, non-functional, and security requirements of the **Academic Performance Analytics System**.

The system aims to provide educational institutions with a centralized platform for tracking student academic performance, analyzing grades, attendance, and assignment submissions, and generating predictive insights about academic outcomes.

This document serves as a reference for developers, testers, project stakeholders, and academic evaluators throughout the software development lifecycle.

1.2 Scope

The Academic Performance Analytics System is designed to help educational institutions monitor, analyze, and report student academic performance.

The proposed system will:

- Maintain student academic records, including grades, attendance, and assignment submissions.
- Allow authorized faculty members to enter and update academic information.
- Calculate performance statistics and identify academic trends.
- Provide visual dashboards displaying academic performance indicators.
- Generate predictive insights about potential student academic outcomes.
- Identify students who may require additional academic support.
- Generate performance reports for faculty and academic administrators.
- Restrict access to sensitive student information based on user roles.

The system will support informed academic decision-making through data analysis, visualization, and reporting.

1.3 Intended Audience

This document is intended for:

- **Developers:** To understand the system requirements and implement the proposed functionality.
- **Testers:** To design and execute test cases based on specified requirements.
- **Faculty Members:** To understand how academic data will be managed and analyzed.
- **Academic Administrators:** To understand reporting, monitoring, and decision-support capabilities.

- **Project Evaluators:** To assess the completeness and correctness of the proposed system.

1.4 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
APAS	Academic Performance Analytics System
GPA	Grade Point Average
ML	Machine Learning
RBAC	Role-Based Access Control
Dashboard	A visual interface displaying academic performance metrics
Predictive Analytics	Analysis of historical data to estimate possible future outcomes
RTM	Requirements Traceability Matrix

2. Overall description

2.1 Product Perspective

The Academic Performance Analytics System (APAS) is a proposed web-based application designed to centralize student academic records and support data-driven decision-making in educational institutions. It integrates student grades, attendance records, and assignment performance into a unified platform for analysis, visualization, and reporting.

2.2 Major Product Functions (Detailed)

- User authentication and role-based access control.
- Student profile and academic record management.
- Grade and assessment data management.
- Attendance tracking and percentage calculation.
- Assignment submission and marks tracking.
- Academic performance analysis and statistical summaries.
- Interactive dashboards and performance visualization.
- Predictive analysis of student academic outcomes.
- Identification of academically at-risk students.
- Generation and export of academic performance reports.

2.3 User Roles and Characteristics (Expanded)

- **Administrator:** Manages user accounts, student profiles, access permissions, and system configuration.

- **Faculty Member:** Records grades, attendance, and assignment information; monitors student performance; and reviews academic analytics.
- **Student:** Views personal academic records, attendance statistics, grades, and performance reports.

2.4 Operating Environment

The proposed system will operate through a web browser on desktop or laptop computers. It will require a web application server, a database for academic records, and an analytics component for statistical and predictive processing. Users will access the application through a network connection. The specific programming languages, frameworks, database technology, and deployment platform will be finalized during system design.

2.5 Constraints

- Access to student information must be restricted according to user roles.
- Predictive accuracy depends on the quality and availability of historical academic data.
- The system must protect confidential student information.
- The application must handle incomplete or invalid academic records appropriately.
- System performance will depend on available infrastructure and network connectivity.

3. External interface requirements

3.1 User Interfaces

The Academic Performance Analytics System shall provide a web-based graphical user interface accessible through standard web browsers.

- **Login Interface:** Allows authorized users to log in using their credentials.
- **Administrator Dashboard:** Provides options for managing users, student records, and system configurations.
- **Faculty Dashboard:** Allows faculty members to enter and update grades, attendance, and assignment records and view academic analytics.
- **Student Dashboard:** Allows students to view their grades, attendance percentages, academic progress, and performance reports.
- **Analytics Dashboard:** Displays performance trends using charts, graphs, tables, and statistical summaries.
- **Reporting Interface:** Enables authorized users to generate and download academic performance reports.
- The interface shall provide clear navigation, appropriate input validation, and understandable error messages.

3.2 Hardware Interfaces

The proposed system does not require specialized hardware. It shall operate using standard computing devices.

- **Client Devices:** Desktop computers and laptops with a web browser.
- **Server Infrastructure:** A computing environment capable of hosting the application, database, and analytics services.
- **Input Devices:** Keyboard and mouse for data entry and navigation.
- **Display Devices:** Monitors for viewing dashboards, reports, and visualizations.

3.3 Software Interfaces

The system shall interact with the following software components:

- **Database Management System:** Stores student profiles, grades, attendance, assignment records, and generated analytics.
- **Web Application Backend:** Processes user requests, enforces access permissions, and manages application logic.
- **Analytics Module:** Performs statistical calculations and academic performance analysis.
- **Machine Learning Module:** Generates predictive insights based on available historical academic data.
- **Reporting Module:** Produces academic performance summaries and downloadable reports.
- **Authentication Service:** Validates user credentials and manages authorized access.

The specific technologies and frameworks will be finalized during the design phase.

3.4 Communications

- The system shall support communication between client browsers, application services, and the database.
- Client-server communication shall use HTTP or HTTPS, with HTTPS required for deployed environments handling student information.
- Data exchanged between frontend and backend services may use JSON-based APIs.
- The system shall validate incoming requests and restrict unauthorized access.
- Communication failures shall produce appropriate error messages without exposing sensitive system information.

- Database connections shall be protected using appropriate authentication and access controls.

4. System features (detailed)

4.1 User Authentication and Access Control

Description: This feature allows administrators, faculty members, and students to securely access the Academic Performance Analytics System according to their assigned roles.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
APAS-F-001	The system shall authenticate registered users using valid login credentials.	Functional	High	Administrator, Faculty, Student	Valid credentials grant access; invalid credentials are rejected. Test: TC-AUTH-01	Requires user database and authentication service.
APAS-F-002	The system shall provide role-based access control for administrators, faculty members, and students.	Functional	High	Administrator, Security	Users can access only functions permitted for their assigned roles. Test: TC-AUTH-02	Depends on role and permission configuration.
APAS-F-003	The system shall allow authenticated users to securely log out and terminate their active sessions.	Functional	High	All Users	After logout, protected pages cannot be accessed without reauthentication. Test: TC-AUTH-03	Requires session mana

4.2 Student Academic Record Management

Description: This feature enables authorized users to maintain student profiles, record academic grades, and manage attendance information for accurate academic performance tracking.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
APAS-F-004	The system shall allow administrators to create, view, update, and manage student academic profiles.	Functional	High	Administrator	Authorized administrators can create and update student profiles, and changes are stored correctly. Test: TC-REC-01	Requires student database and administrator access.
APAS-F-005	The system shall allow authorized faculty members to enter and update student grades for assessments and examinations.	Functional	High	Faculty	Valid grades are saved and displayed correctly; invalid grade values are rejected. Test: TC-REC-02	Depends on student profiles and assessment records.
APAS-F-006	The system shall allow authorized faculty members to record student attendance and automatically calculate attendance percentages.	Functional	High	Faculty, Student	Attendance records are saved, and the calculated percentage matches the recorded attendance data. Test: TC-REC-03	Requires attendance records and calculation logic.

4.3 Academic Performance Analysis

Description: This feature enables the system to analyze student grades, attendance, and assessment results to evaluate academic performance and identify areas requiring improvement

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
APAS-F-007	The system shall automatically calculate students' overall academic performance using recorded grades and assessment scores.	Functional	High	Faculty, Administrator	The calculated performance values match the assessment data and configured calculation rules. Test: TC-ANA-01	Depends on student grade and assessment records.
APAS-F-008	The system shall analyze students' academic performance across different subjects and semesters to identify performance trends.	Functional	High	Faculty, Student	The system displays accurate subject-wise and semester-wise performance comparisons. Test: TC-ANA-02	Requires historical academic records.
APAS-F-009	The system shall identify students with low academic performance or attendance below configured thresholds.	Functional	High	Faculty, Administrator	Students meeting the configured risk conditions are correctly identified and flagged. Test: TC-ANA-03	Depends on grade analysis, attendance records, and configurable thresholds.

4.4 Academic Performance Visualization and Reporting

Description: This feature provides interactive dashboards, graphical representations, and downloadable reports to help students, faculty members, and administrators understand academic performance

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
APAS-F-010	The system shall provide an interactive dashboard displaying student grades, attendance percentages, and overall academic performance.	Functional	High	Faculty, Student, Administrator	Dashboard displays accurate academic data corresponding to the selected student and authorized user. Test: TC-VIS-01	Depends on academic records and performance calculations.
APAS-F-011	The system shall generate graphical visualizations, including bar charts and line graphs, to represent academic performance trends.	Functional	Medium	Faculty, Administrator	Charts accurately represent stored academic data and calculated performance metrics. Test: TC-VIS-02	Requires analytics module and visualization components.
APAS-F-012	The system shall allow authorized users to generate and download academic performance reports containing grades, attendance statistics, and performance summaries.	Functional	High	Faculty, Administrator	Reports are generated successfully, contain accurate academic information, and can be downloaded. Test: TC-REP-01	Depends on academic records, analytics module, and reporting servic

4.5 Predictive Analytics and Academic Risk Identification

Description: This feature enables the system to predict students' academic outcomes using historical performance data and identify students who may require additional academic support

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
APAS-F-013	The system shall use historical academic records, including grades and attendance, to generate predictions of students' future academic performance.	Functional	High	Faculty, Administrator	The system generates predictions for students with sufficient historical data and displays the predicted outcomes. Test: TC-PRED-01	Requires historical academic data and a trained prediction model.
APAS-F-014	The system shall identify students at risk of poor academic outcomes based on predicted performance and predefined risk criteria.	Functional	High	Faculty, Administrator	Students meeting the configured risk criteria are correctly classified and displayed in an academic risk list. Test: TC-PRED-02	Depends on prediction results and configured risk thresholds.
APAS-F-015	The system shall provide academic performance insights and recommendations based on identified performance trends and risk factors.	Functional	Medium	Faculty, Student	The system displays recommendations relevant to identified academic weaknesses and risk factors. Test: TC-PRED-03	Depends on performance analytics and recommendation rules.

5. Non-functional requirements (detailed)

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
APAS-NF-001	The system shall load academic dashboards within 3 seconds for at least 95% of requests under normal operating conditions.	Performance	High	95th percentile response time ≤ 3 seconds. Test: TC-PERF-01
APAS-NF-002	The system shall maintain at least 99% monthly availability, excluding scheduled maintenance.	Reliability	High	Monthly uptime reports confirm availability ≥ 99%. Test: TC-REL-01
APAS-NF-003	The system shall restrict access to academic records based on authenticated user roles and permissions.	Security	High	Unauthorized access attempts are rejected. Test: TC-SEC-01
APAS-NF-004	The system shall maintain timestamped audit logs of academic record modifications and retain them for at least 12 months.	Audit/Data Retention	High	Audit logs contain user ID, timestamp, and modification details and can be retrieved for the required retention period. Test: TC-AUD-01
APAS-NF-005	The system shall provide an accessible user interface supporting keyboard navigation and sufficient text contrast in accordance with applicable WCAG 2.1 AA criteria.	Usability/Accessibility	Medium	Accessibility audit confirms compliance with the applicable WCAG 2.1 AA criteria. Test: TC-UX-01

5.1. Security

5.1.1 Security Objectives

Objective ID	Security Objective	Description
APAS-SO-001	Confidentiality of Academic Records	Ensure that student grades, attendance, personal information, and academic performance reports are accessible only to authorized users based on their assigned roles.
APAS-SO-002	Integrity and Accountability of Academic Data	Prevent unauthorized modification of academic records and maintain audit logs to track changes made by authorized users.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
APAS-SR-001	The system shall require users to authenticate using valid credentials before accessing protected academic information.	Security	High	Valid credentials allow login; invalid credentials are rejected. Test: TC-SEC-01
APAS-SR-002	The system shall enforce role-based access control for administrators, faculty members, and students.	Security	High	Users can access only the functions and records permitted for their roles. Test: TC-SEC-02
APAS-SR-003	The system shall protect sensitive academic information during transmission using HTTPS with TLS 1.2 or higher.	Security	High	Network connections use TLS 1.2 or higher; insecure connections are rejected or redirected. Test: TC-SEC-03
APAS-SR-004	The system shall securely hash user passwords using an approved password-hashing algorithm before storing them.	Security	High	Passwords are not stored in plaintext, and password verification works correctly. Test: TC-SEC-04

APAS-SR-005	The system shall record security-relevant events, including login attempts and academic record modifications, in timestamped audit logs.	Security	High	Audit logs correctly record user identification, event type, and timestamp. Test: TC-SEC-05

6. Quality attributes & Acceptance tests

Acceptance Criteria:

- All high-priority functional requirements (APAS-F-001 to APAS-F-015) shall be implemented and verified against their defined acceptance criteria.
- All five non-functional requirements (APAS-NF-001 to APAS-NF-005) shall be tested against their specified performance, reliability, security, auditability, and accessibility criteria.
- All five security requirements (APAS-SR-001 to APAS-SR-005) shall pass their corresponding security test cases.
- The system shall accurately calculate grades, attendance percentages, and academic performance metrics using the stored student records.
- The predictive analytics module shall be evaluated using appropriate test data, and its performance results shall be documented.
- No unresolved critical-severity defects shall remain at the time of final acceptance.

Acceptance Test Suites:

- Authentication and Role-Based Access Control Tests
- Student Academic Record Management Tests
- Academic Performance Analysis Tests
- Dashboard Visualization and Reporting Tests
- Predictive Analytics and Academic Risk Identification Tests
- Performance, Reliability, Security, Audit Logging, and Accessibility Tests

7. System models and diagrams

7.1 UML Use-Case diagram

The Academic Performance Analytics System (APAS) is represented using two UML use-case diagrams. The first diagram illustrates academic record management functionalities, while the second illustrates performance analytics, prediction, and reporting functionalities. These diagrams identify the interactions between administrators, faculty members, students, and the system.

Diagram 1 -- Academic Record Management.

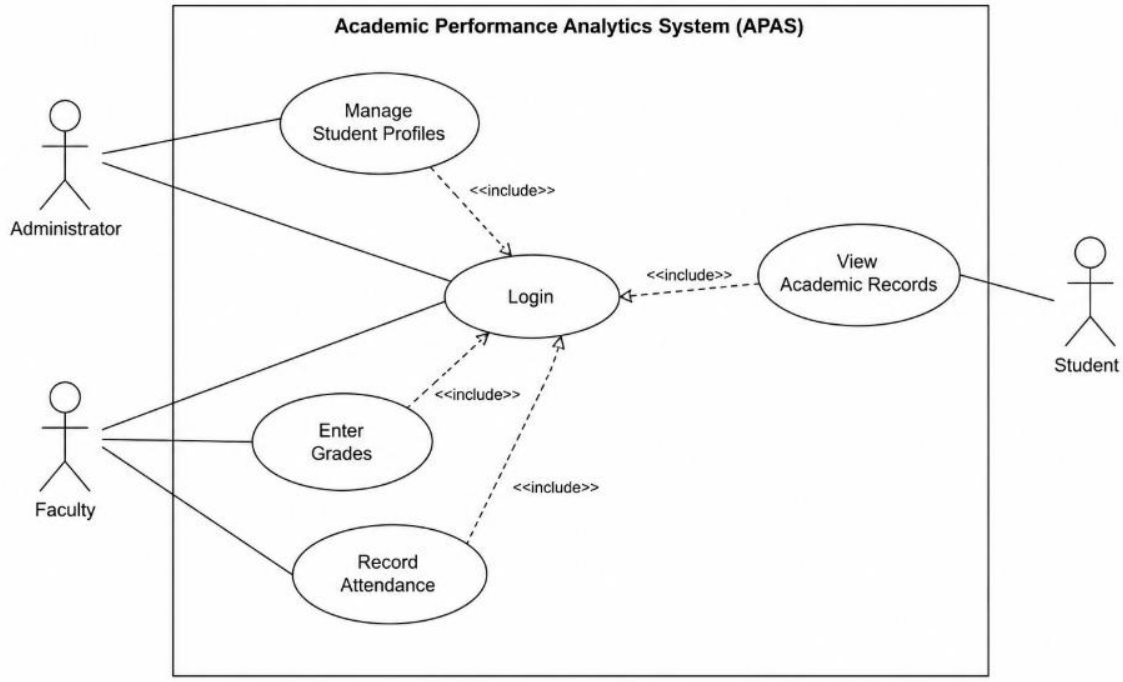
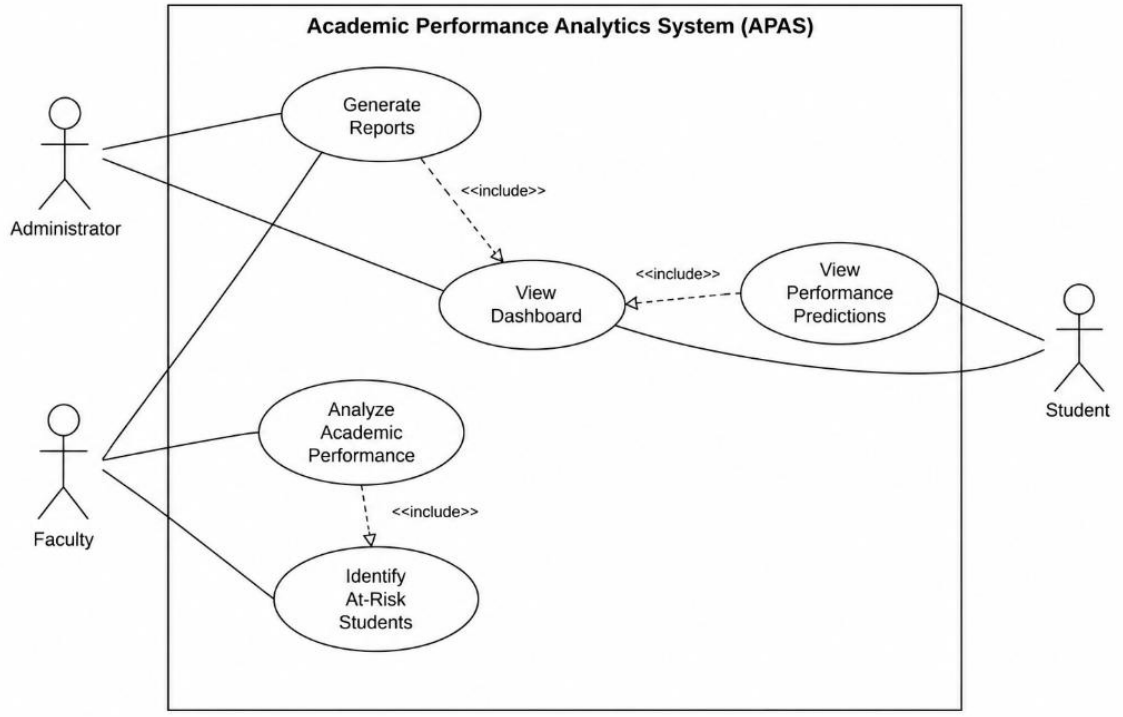


Diagram 2 — Performance Analytics and Reporting



8.Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix (RTM) maps the functional, non-functional, and security requirements of the Academic Performance Analytics System (APAS) to their corresponding test cases. It ensures that all specified requirements are covered during software testing and validation.

Requirement ID	Requirement Description	Requirement Type	Priority	Test Case ID
APAS-F-001	User authentication and login	Functional	High	TC-AUTH-01
APAS-F-002	Role-based access to system functions	Functional	High	TC-AUTH-02
APAS-F-003	Secure Logout and Session Termination	Functional	High	TC-AUTH-03
APAS-F-004	Student academic profile management	Functional	High	TC-REC-01
APAS-F-005	Student grades entry and updating	Functional	High	TC-REC-02
APAS-F-006	Attendance recording and percentage calculation	Functional	High	TC-REC-03
APAS-F-007	Overall academic performance calculation	Functional	High	TC-ANA-01
APAS-F-008	Subject-wise and semester-wise performance analysis	Functional	High	TC-ANA-02
APAS-F-009	Identification of students below performance or attendance thresholds	Functional	High	TC-ANA-03
APAS-F-010	Academic performance dashboard	Functional	High	TC-VIS-01
APAS-F-011	Graphical visualization of academic trends	Functional	Medium	TC-VIS-02
APAS-F-012	Academic performance report generation	Functional	High	TC-REP-01
APAS-F-013	Prediction of future academic performance	Functional	High	TC-PRED-01

APAS-F-014	Identification of academically at-risk students	Functional	High	TC-PRED-02
APAS-F-015	Academic insights and recommendations	Functional	Medium	TC-PRED-03
APAS-NF-001	Dashboard response time within specified limits	Non-Functional	High	TC-PERF-01
APAS-NF-002	Monthly system availability	Non-Functional	High	TC-REL-01
APAS-NF-003	Access restrictions for academic records	Non-Functional	High	TC-SEC-01
APAS-NF-004	Audit logging and data retention	Non-Functional	High	TC-AUD-01
APAS-NF-005	User interface accessibility	Non-Functional	Medium	TC-UX-01
APAS-SR-001	Secure user authentication	Security	High	TC-SEC-01
APAS-SR-002	Role-based access control	Security	High	TC-SEC-02
APAS-SR-003	Secure data transmission using TLS	Security	High	TC-SEC-03
APAS-SR-004	Secure password hashing	Security	High	TC-SEC-04
APAS-SR-005	Security event audit logging	Security	High	TC-SEC-05